

INTELLECTUAL PROPERTY LAW II (PATENTS)

Intellectual property rights are rights given to a person over the creation of the minds, these rights always give the creator authority over the use of his or her creation for a certain period of time.

While there is no official definition of IP Article 1(2) of the Trips agreement provides that for the purposes of the agreement, intellectual rights refer to all categories of rights which are covered by Article 1-7 of part 2 of the Trips agreement. They include copy right and related right, trademarks and service marks, geographical indications, industrial designs, patents, layout ground for integrated circuits and protection of undisclosed information.

TRIPS AGREEMENT

The general goals of the agreement are contained in the preamble of the agreement. The agreement provides some of its objectives to include the reductions to and impediments to international trade, promotion of effective and protection of I.P rights and ensuring that measures and procedures to enforce IP do not themselves become barriers to legitimate trade.

The objectives of the TRIPS agreement should be read in conjunction with Article 7 which provides that the protection and enforcement of the IP rights should contribute to the promotion of the technological innovations and the transfer and dissemination of technology when mutual advantage of the producers and the users of technological knowledge in a manner conducive to the social and economic welfare at the balance of rights and obligations.

In addition, *Article 8 of the TRIPS Agreement* provides the guiding principles for the formulation and the amendment of the national laws and regulation. This Article provides that members are at liberty to adopt measures necessary to protect public

health and nutrition and promote public interest in sectors of vital importance for their social, economic and technological development provided that the measures are consistent with the provisions of the Agreement. The Agreement further states that these measures should be taken to avoid the abuse of IP rights by the rights



owners including doing away with practices which may unreasonable restrain and international transfer of knowledge.

The agreement has five broad areas;

The basic principles of trading systems and other IP

agreements Giving adequate protection of the IP rights

Guidance on how countries should enforce IP rights in their

territory Settlement of disputes of IP rights between members

of WTO

Special transitional arrangements for the periods in which the new system would be introduced.

PRINCIPLES OF NATIONAL TREATMENT

This principle is provided for under Article 3 of the agreement. It requires that members should uphold the same treatment provided to its nationals and to other members of WTO. It is important to note that the agreement provides a number of exceptions under the relevant conventions and in respect to performers, producers of programs and broadcasting organizations as provided for under Article 3.

THE MOST FAVOURED NATION PRINCIPLE

Article 4 of the agreement contains the principle of the most favored nation which has not been traditionally provided for in the context of IP and at multilateral level. This principle provides that any advantage, favor, privilege or immunity provided by a member to nationals of any country whether a member of WTO or not, shall be accorded immediately and unconditionally to the nationals of all other members. As in the case with national treatment, procedures provided in the multilateral agreement concluded under the auspices of WTO relating to the acquisition or maintenance of I.P is exempted from this principle.

PATENT LAW

A patent is an exclusive right granted for an invention which is a product or a process that provides in general a new way of doing something or offer a new technical solution to the problem, see ***section 2 IPA 2014***. In order to be patentable, must be of practical use, it must show an element of novelty meaning it must have some new characteristic which is not known in the



body of the existing knowledge of that particular field which would be called prior art. The invention must also show an inventive step which could not be known by a person with average knowledge of the technical field, see **section 9 IPA**. It is also important that the subject matter must be accepted as patentable under the law of a given nation. See **section 8(2) IPA**

Once the patent expires, its protection ends and the invention in question is said to have entered public domain meaning that the owner no longer holds exclusive rights to the invention and it becomes available to commercial exploitation by others.

Patents provide incentives to individuals by offering them recognition for their creativity and material reward for their inventions. These incentives encourage innovations which ensure that the quality of human life is continually enhanced. All the patent owners have an obligation in return for patent protection to publically disclose information on their innovations in order to enrich the body of technical knowledge that exists as such an ever increasing knowledge of public promotes further creativity and innovation. In this way the patents not only provide protection for the owner but also valuable information and inspiration for the future generation.

HOW ARE PATENTS GRANTED AND WHO GRANTS IT?

The first step in securing a patent is the filing of a patent application. The patent application generally contains the title of the invention as well as indication of its technical field. You must include the background and in a clear language and enough detail that an individual with an average understanding of the field could use and reproduce the invention. Such descriptions are usually accompanied by usual materials like drawings, plans for better describing the invention.

In general, where no patent exists, an application for a patent must be filled and the patent shall be granted and enforced in each country in which you seek patent protection in accordance with the laws of that country.

TYPES OF PATENT PROTECTION

Patent protection can be under 3 types of categories

- Utility patents
- Design patents
- Plant patents



UTILITY PATENTS

A utility patent is a patent that covers the creation of a new or improved and useful product, process, or machine. A utility patent, also known as a "patent for invention," prohibits other individuals or companies from making, using, or selling the invention without authorization. When most people refer to a patent, they are most likely referring to a utility patent.

DESIGN PATENT

These apply new, original and ornamental designs or an article or manufacture. The major difference between design and utility patents can be seen in today's computers and phones. Under this, the plastic shell that covers all the working parts is covered by a design patent while all the many working designs it has are covered by utility patents.

PLANT PATENTS

These are granted by any person who has invented or discovered and a sexually reproduced any distinct and new variety of plant including cultivated hybrid and newly found seedlings (protection of Plant varieties see Article 27 (3)); by patent, by an effective sui generis system, any combination thereof.

Patents are provided for under Article 27 in the TRIPS agreement. This article requires that for a patent to be granted it should be for an invention that is either a product or process in all fields of technology provided that the process is new, involves an inventive step and is capable of industrial application.

The article provides that member states may exclude invention from patentability on a number of grounds including where it important to protect a public order or morality & protection of human, animal and plant life and invention that may be pre-judicial to the environment.

The article further excludes from patentability, the diagnostic & surgical methods of treatment of animals & humans and also recognizes that plant varieties could be

protected under patents & effective sui-generis system on a combination of the law.

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Utility Models and Patents

Utility Models

A Utility Model is an exclusive right granted by the government for an innovation/invention, which is either a product or process that offers a new technical solution to a problem. A product or process that is new and is useful can be protected using this system. Their term of protection is 10 years and is not renewable. ***See section 69(3) IPA.*** Registration for a utility model is simple and fast, and gives the holder the right to exclude others from exploiting the protected innovation/invention.

Utility Models provide protection for incremental improvements to products and processes and it is very relevant for SMEs. For Example: Printing- roller cleaning system, fruit sorting machine, simple bottle cleaning machine etc.

Patents

A patent, like a utility model, is an exclusive right granted by the government for an invention. Protected inventions can range from simple things like a safety pin to sophisticated items like juice processing machine.

An invention that is Novel, Inventive and is industrially applicable may be granted using a Patent. The term of protection for patents is 20 years. For the patent to remain in force the patent holder is required to pay annual maintenance fees.

Requirements for protection explained

- Novel /New– The invention/innovation should not be disclosed in any publication anywhere.
- Inventive step- The new product or process should not be obvious to a person skilled in the art.
- Industrial Application- The invention can be made or used in some kind of industry

Rights of a Utility Model/Patent Holder

Decide who may or may not exploit the protected invention.

Permit/license the use of the invention on mutually agreed terms or sell the invention outright.

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THE CONCEPT OF NOVELTY

The novelty test determines if the subject matter the invention has been previously known by others. It is important to note that novelty is not something that can be proved or established but it is rather something that can be proved through determining its absence. This means that the key issues under novelty always revolve around establishing its absence. Key to the novelty test is the concept of prior art. This refers to the existing body of knowledge either written or oral that governs the planned subject matter of the invention. In this light, an invention is deemed novel (new) if it's not predated by prior art i.e. if prior to the filing of the patent application, an invention does not already exist as part of the body of knowledge known in the planned subject matter.

Article 27 of the TRIPS agreement recognizes novelty as one of the key aspects for patentability of an invention. It is important however to note that the TRIPS agreement does not make an effort to unpack what the concept of novelty should mean for the member states as such, member states have the liberty to determine what constitutes novelty in their national legislation of patents. In Uganda the patent Act addresses the aspect of novelty under *section 10 IPA*.

This *section 10* emphasizes that an invention is new if it is not anticipated by prior art. The section further defines prior art everything made available for the public anywhere in the world by means of written disclosure including drawings and other illustrations or by oral disclosure including through use and any other non-written means. A couple of exceptions need to be considered as provided under *section 10 (2)*. The Uganda law also mentions that disclosure to the place of the invention should not be taken into consideration if this occurred within 12 months preceding the date on which the application was filed and this should be by reason of consequence of either acts committed by the or his or her predecessor or by an abuse committed in relation to the applicant or his or predecessor in the title.

As such it is important to note that prior art is lost if disclosure is proved and this

may be under three main instances;

Printed publication-under this description of an invention may be published in a writing or in a publication but this must be in a tangible format and they are in most cases be a physical carrier of the information which makes the subject available to the public.



Oral disclosure-the description of the invention in words spoken in place (not necessarily recorded) will also amount to prior art of that invention in question. This may include lectures and radio broadcasts

Disclosure by use-this includes the use of the invention in public or putting the public in a position that enables them to know about the invention.

The common type of usual disclosure includes putting the invention on sale or display, public demonstrations, unrecorded television broadcast and an actual public use. For document to destroy the novelty of an invention it must clearly and fully describe the subject matter described in the document.

The invention is then found not to be novel if the document involves all the characters of the claim in question and in that case the document is said to have anticipated the subject matter of the pending application and the existence of prior art is claimed.

It should also be noted that when considering novelty, it possible to combine separate items with prior art together. In the case of *Windsurfing international Inc v Tabur marine [1985] RPC 59* it was highlighted that a product which preceded the patent would infringe or affect the right of the patent if it surfaced at a later date. In this particular case, prior art was deemed to have happened under the act of the use with the publication. The case points out that its only public information that and taken into account but no matter it is substituted and in what language it is written as long as it discloses the invention it will destroy the patentability of the invention in question.

The parameter for determining novelty would seem to be fairly objective as long as the invention has been made available to the public. This was also interpreted *in GENETECH-INC'S PATENT (1989) RPC 147* where it was noted to form part of the state of the art, the information given by the user must have been available by at least one member of the public who was free in law to make use of it. The implication of this judicial interpretation is that if the information to a person or group of persons

under circumstances which make it to disclose to any other person or to make use of the invention cannot be said to having not been available to the public as to form part of the state of the art. It has also been judicially interpreted that to form part of the state of the art the disclosure of the invention must be an enabling disclosure and must have provided



sufficient information to enable the person skilled in that art to make use of it. The courts have also tended to be very willing to declare that an invention has been made to the public where there is a possibility that the particular person has further disclosed the information to more than one person, for instance it has been held that if the invention was disclosed by a book which has not been sold but only displayed only in a book shop.

It was stated that sufficient disclosure had been made to make the invention part of the state of the art. Similarly, where a book is written in French and it is in a British museum in a room not accessible to the public but its title on summary is included on the catalog, it has been held that it had been made available to the public.

In the case of *Merrell Pharmaceuticals V Norghthon* it was noted and held that prior use of a product was to be considered in the same way as prior use published document but in both cases prior use will only invalidate where the information made available will enable the person skilled in the art to work the information to make the invention.

INVENTIVE STEP OR NON OBVIOUSNESS OF AN INVENTION

One of the most complex aspects of patent law is the determination of inventive step or non- obviousness of an invention. Inventive step is judged through a line of thought that a man is skilled in the art but lacks in the inventive genius. A skilled person is one having all the standard knowledge available in the field and having the standard capabilities of routine work and experimentation allowing him to straight forward progress from what is already known. Such a person nearly has a sense of what is possible but lack the imagination/inventiveness and has no benefit of foresight beyond the available knowledge.

The question to be asked is whether this normal skilled person would have used his mind to make a breakthrough in the alleged invention. This is a critical step in the requirement. For one to be able to move beyond a new innovation to something that is not obvious to a person skilled in the art. The trips agreement under **Article 27 (1)**

provides that for an invention the subject of patent should involve an inventive step.

According to *section 11 of the Industrial Property Act 2014*, an invention is said to have an inventive step if having regard to prior art as defined in **section 10(3)**, it would not have been



obvious to a person skilled in the art, on the date of filing of the application and if priority is validly claimed, the time considered is the priority date.

It is important to keep in mind the essence of an invention, which includes the identification of a problem, creation of a solution to that problem and guarantee that the result will be positive in applying the solution created.

If the problem is known/considered obvious, then the inventive step of the solution is examined. If there is no originality found in the solution, then the assessment passes on to the result to determine if the result is obvious or whether it is surprising either by its nature or its existence. Thus, there lacks an inventive step if a person with the ordinary skill in the given art is unable to pose a problem, solve the problem and foresee the results.

In ***Hotchkiss v Greenwood*** 52 US

Court noted that unless more ingenuity and skill in applying the old method of fastening the knob were required in the application of the same to the clay knob than were possessed by an ordinary mechanic acquainted with the business, there was an absence of the degree of skill and ingenuity that constitute essential elements of the invention. In other words, the improvement in the work of a skilled mechanic and not an inventor is required to pass the test of non-obviousness.

This case emphasized the fact that patents would only be granted to something that is a novel invention and not just a minor improvement in the existing knowledge.

In ***Graham v John Deere company***

Court found that patents were only intended for those inventions which were new, useful, furthered human knowledge rather than for small and details and obvious improvements. Courts found the following as the factors for determining non-obviousness;

Scope and content of prior Art

Differences between the claimed invention and prior

art Level of ordinary skill in the prior art

Secondary considerations e.g. commercial success, unresolved needs, failure of others.



Court found that Graham's invention served the same purpose as the 1st one and technically the improvement was found to be obvious by anyone who read Graham's patent.

INDUSTRIAL APPLICATION

Industrial application refers to the usefulness of the invention in question. It is not every invention which is new and involves an inventive step could be patentable. This is because patents are designed to promote industrial development and they provide incentives for creativity for people involved in the research and industrial development endeavors. Just like novelty and inventive step, industrial application is required by Article 27(1) of TRIPS Agreement. This article defines Industrial application to be synonymous with the term usefulness. As such, countries are at liberty to determine what is useful before being patented at the national level.

Section 12 of the Industrial Property Act 2014, Uganda considers Industrial application to include inventions that are considered industrially applicable by their nature and can be technologically made, used in any industry.

In *Lowell v Lewis*

Court found for Lewis and held that to warrant a patent, the invention must be useful. Court defines usefulness as something that is capable of some beneficial use, in contradiction to what is frivolous or worthless. According to court, usefulness does not mean better, it just means different. It pointed out that a new invention to poison people or promote immorality, facilitate private assassinations may not be a patentable invention. If the invention steers wider of these objections, whether it be more or less useful, is a circumstance very immaterial to the interests of the patentee, but of no importance to the public. Basically, even if the patent was not as good as the other pumps on the market, it was still patentable. Court noted that it is extremely hard to judge what is better, as an invention might be worse than most things, but better for

one specific thing.

PROPERTY RIGHTS AND EXPLOITATION

The right to grant a patent

A patent can be granted to the following categories



The inventor

The co-

inventor

The employer of the inventor when the invention is made during the course of employment

EMPLOYEE INVENTIONS

THE RIGHT TO GRANT EMPLOYEES

“If people don't get paid for their inventions, that's not a good thing. In the case of many patents, there are people who aren't in a position to take them to the next level. If you don't enforce your rights, no one is going to enforce them for you.”-Nathan Myhrvold, former Chief Technology Officer at Microsoft, co-founder of Intellectual Ventures and the principal author of Modernist Cuisine.

Today all over the world a great number of Patent grant applications are by inventors who are employed in a given corporation or company. Most of the employees that is; persons who work or worked under contract of employment make these inventions while employed under a contract of employment.

The question has always been “who owns the right to acquire a patent? Employer or employee and if it is employer who owns the patent is there any level of protection afforded to the deviser of the patent to benefit from his/her ideas?

Background (UK Patents Laws on Employee Inventors)

The general rule is that an employee invention belongs to the employee unless; the employee signed an agreement assigning the invention rights or was specifically hired for his/her invention skills or to create invention. Under such circumstance then the employer owns the invention and in the absence of an agreement fitting within the above mentioned, other means have been used to determine ownership by the employer.

Under section 39 of U K's Patent's Act of 1977 an invention made by the employee shall be taken to belong to his/her employer. This situation nevertheless, might any case as already stated above be expressly covered by the contract of employment. Various scholars most notably



Bainbridge in the book Intellectual Property, have labored to lay down the two main circumstances an employer may claim such an invention as of right and they are as follows

1) Where an invention has been made during the course of employees' normal duties as an employee or in the course of duties specifically assigned to him/her provided that the invention can under reasonable circumstance be attributed to carrying out of those duties.

2) Where the employee at the time of the invention was such that he /she had a special obligation to further the interest of the employers undertaking.

It should be noted that ascertaining the meaning of "normal" course of duty of the employee has been narrowed by the court to avoid vague interpretation that may arise. In the case of *LIFFE Administration and Management v Pavel Pinkava (2007) RPC 667* court stated that it was not acceptable to ascertain the meaning of "normal" by reference to some other standards such as "ordinary", "day to day" or primary.

Where the invention cannot be deduced from the above two circumstances, it can then be said without fear of contradiction that the invention may as belong to the employee who reserves all the rights that accrues thereunder.

The above circumstances seem to have given more protection to the employee whose invention may by right be claimed by the employer under the guise that it was done during the normal course of the employee's duties.

But the employer may still acquire a limited right to use the employees patent without paying. Courts have however, in various cases held that simply because an individual is employed does not necessarily grant the employer ownership of the patent. Each case presents different facts and must be judged basing on such as per Bainbridge.

This is because a patent is a form of personal property; it may be transferred, created or granted only in accordance with the law. The proprietor may offer to surrender it anytime to any person including his employer.

A Patent is always granted to the inventor or joint inventors, except where someone else has a better entitlement to it under the law. In most cases (especially in employment situations) the inventor may not be the proprietor/owner of a patent but in some the inventor would be the



proprietor. It is trite law (under Section 39 of The Patents Act (UK), that the inventor will always be the deviser of the invention even without ownership or protection of any sort.

The law provides that the proprietor is the person to whom the patent has to be granted and where the proprietor is not the inventor, he/she has a right to be mentioned as being the inventor in any patent granted and in any published application. (Bainbridge)

Section 40 of the same Act above, provides for compensation of employees who are actual deviser, whose invention is of outstanding importance to the employer and the patent has been registered.

Uganda's Current Situation

In Uganda invention made by an employee is enshrined in the Industrial Properties Act 2014 under **section 19**. It provides that in the absence of a contract to the contrary, the right to a patent for an invention made in execution of a contract of employment belongs to the employer. However, the Act under **Section 19(2)**, further states that, where the invention is of exceptional importance the employee has a right to equitable remuneration taking into consideration his or her salary and the benefit derived by the employer from the invention.

It should be noted with concern that **section 19(1)** of the Industrial Properties Act grants patent rights to employer in absence of a contract to the contrary where a contract of employment does not require the employee to exercise any inventive activity but when the employee has made the invention by using data or means available to him or her during his or her employment (**section 19(2)**).

And where such an invention has been of special importance then the employee as of right is entitled to remuneration taking into consideration his or her salary and the benefit derived by the employer from the invention (**section 19(2) (4)**). This in plain terms despite being a right to an employee places the power to determine how special an invention is on the employer which may be detrimental to the employee in the

long run despite court having a final say in the absence of an agreement (*section 19(5)*).

The law however provides that an invention made without any relation to an employment or contract of service and without the use of the employer's resources, data, means, materials,



installations or equipment belongs solely to the employee or the person commissioned as seen in

Electrolux V Hudson and Others (1977).

General Considerations on ownership of Employee patent

Issues of ownership rarely arise where the employee signed employment agreement or a pre- invention assignment or where one is employed to invent. If a person's employment agreement contains a pre invention assignment, then any invention made by the employee does not belong to him or her.

Also under the doctrine of employed to invent courts have held that “any one employed to make inventions, who succeeds during his term of service, in accomplishing that task is bound to assign to his employer any patent obtained.” These situations are not governed by **Section 19** of Industrial Properties Act which is only concerned with situations where there are no agreements.

The protection accorded to employee under Industrial Property Act as already seen earlier is very minimal despite the rights of patent belonging to the inventor under **Section 17** and **Section 38** of the *Industrial Properties Act 2014*, which outlines the rights of the patent owner to be exclusive of others and free from interference.

In some cases, this protection may mainly be offered to the lower level workers and not the directors as illustrated in the case ***Christopher S. French v Paul J Mason (1999) FSR 597*** where court held that directors of a company may not have any right to protection or entitlement to patent. This is because they are high level employees whose nature of work is not clearly defined since they generally manage all the affairs of the company.

It has also been argued that in the absence of a term or a condition in the contract does not prevent an employer being able to claim a patent for an invention devised by an employee. Courts will carefully consider the duties of the employee irrespective of the place where the invention took place. This was the case in ***Re Harris's Patent 1(1985)*** where it was stated that it is necessary to decide what constitutes inventing.

Normal duties or alternatively what duties lying outside that category that were 'specifically assigned to him is also important in determining who owns the patent.



Despite minimal protection under the Act, protection and ownership of patents or inventions by employees is afforded under **section 19(6)** which states that an invention made without any relation to an employment or contract of service and without the use of the employer's resources, data, means, materials, installations or equipment belongs solely to the employee or the person commissioned.

This is also the same under section 37 of the 1977 Patent Act on UK, cases and common law, the mere fact that the invention was made in the employer's time, using employer's resources, where the employer is knowledgeable about it is necessarily conclusive that patent will be granted to the employer.

In *Mellor v Beardmore (1927)*, the court held that the mere existence of a contract of service in no way disqualifies a servant from taking out a patent in his own name and entirely for his own benefit – and that notwithstanding the fact that he has used his employer's time and materials to aid him in completing his invention, either express or implied, to communicate the benefits of his invention to those who employ him.

An employee may in some special circumstances, with or without express or implied conditions use the resources of the employer to invent and own a patent. It does not matter whether the invention is related to his or her duties as long the relevant authorities are satisfied that the description of his or her duties does not involve invention or solving a given problem or is wide.

Section 37 of UK's Patent Act was referred to while determining the right to a patent after grant in Greater Glasgow Health Board's Application 31 to see whether an employed hospital doctor was entitled to be the proprietor of a patent for the invention of a spacing device for an ophthalmoscope. The doctor, a registrar, had a very wide job description which referred to research facilities of which the doctor was encouraged to avail himself for basic clinical research.

The contract did not, however, express this as a duty and the court held that his duty to treat patients did not extend to devising new ways of diagnosing and treating patients. The doctor was entitled to be the proprietor of the patent. He had made the

invention in his own time, during a period when he was working over 80 hours a week (quoted by Bainbridge, Intellectual Property, 8th Ed.)

Compensation



Where an employee successfully invents under a contract of employment and he/she is not a proprietor then the employee has a right to equitable remuneration taking into account his or her salary, the importance of the invention and any benefit derived from the invention by the employer under **section 19(4)** of the Industrial Properties Act 2014.

It is fair and just that where an employee does not own a patent he /she receive a fair share of the employers' benefit as seen in ***Kelly V GE Health Care (2009) EWHC 181***. This is intended to compensate the employee for the ideas of the mind that contributed towards the invention that has proved so important to the employer. Although Cases of compensation are not common however, the first reported case in which an application for employee compensation succeeded is ***Kelly v GE Health care (2009) EWHC 181***.

In the same case Floyd J laid down the principles to follow while granting compensation among them being that the employee must be the actual deviser, the invention must be patented by the employer and it must be of an outstanding importance to the employer

Basically, with the new legislation in place for Patents and limited jurisprudence in Uganda on protection of employee patent protection, the examination of the such in comparison with the UK, it is important to establish the extent of protection accorded to employee inventors.

The *Industrial Properties Act Of 2014* to a great extent grants substantial rights to employers over employee inventors in owning a patent. Employees are mainly protected in term of remunerations in form of compensation which can only be granted where the invention is of outstanding importance to the employer. This may be quite unfair where the employer is deriving benefits from the invention and courts construe less important. The law is therefore lacking in that aspect.

INFRINGEMENT OF PATENTS

Patents confer rights and obligations to patentees. The grant of a patent allows its

owner to exclude others from exploiting the patented invention. The right of the owner is called exclusive because it allows the exclusion of others from exploiting the invention and because the owner is the only one allowed to exploit the invention as long as others are not given an authorization, for example, by way of license to do so. This exclusive right of the patent owner has two main applications in practice, namely protection against infringement and the possibility of assigning



or licensing the right, in part or in whole. The making of the invention in particular, and its development for industrial application, usually involve considerable expense for the applicant and for the future owner of the patent for invention. The patent owner thus wishes to recover this expense through exploitation of the patented invention, in particular through the sale of products that incorporate the invention.

It should be noted that subject to sections 42, 43 and 50, where a person other than the owner of a patent or a licensee does any of the acts specified in **section 38(1)** in respect of the patented invention, that act shall constitute infringement of the patent. Thus, where the owner of a patent or the licensee is of the opinion that his or her patent has been or is about to be infringed, he or she may institute infringement proceedings in the High Court for the following relief; (see section 40 IPA)

(a) Damages

(b) An injunction to prevent infringement or continuing infringement of the patent;

(c) Any other civil remedy.

However, a defendant to an infringement action may argue non-infringement and may seek declarations to that effect. Therefore, one has to be mindful of the likely defenses that may be raised against the infringement action. See rights of a patent holder under section 38 of the Industrial Property Act. Briefly these include

b) (i) when the patent has been granted in respect of a product;

- i. making, importing, offering for sale, selling and using the product;
- ii. stocking the product for the purposes of offering for sale, selling or using the product;

b) (ii) when the patent has been granted in respect of a process—

- i. using the process;
- ii. doing any of the acts referred to in paragraph (a) of this subsection in respect of a product obtained directly by means of the process.

From the foregoing subject to section 42, the owner of the patent has a right to preclude any person from exploiting the patented invention in the manner referred to in subsection (1).



The scope of protection available under the patent is determined by the terms of the claims, but where these are inadequate, the drawings and the description included in the patent may be used to interpret the claims.

What is infringement?

Although the terms “infringe” or “infringement” are used in the Industrial Property Act, case law defines infringement as “any act that interferes with the full enjoyment of the monopoly granted to the patentee” or any activity that deprives the inventor, in whole or in part, directly or indirectly, of full enjoyment of the monopoly conferred by law. According to WIPO an infringement of the exclusive right of a patent owner involves the unauthorized exploitation of the patented invention by a third party.

Who can sue?

The exclusive rights enumerated in **section 38** of the IPA are granted to “the patentee and under **section 42**, a licensee has a right of action. “Patentee” is “the person for the time being entitled to the benefit of a patent”.

-Owner & licensees have locus to sue **section**

50 PA. What constitutes infringement analyzed?

The question of infringement of a patent involves consideration of two matters. The first is whether the article or process alleged to infringe falls within the scope of the monopoly granted by the patent; and the second is whether the alleged infringer has done, in relation to that article or process, any act the monopoly in which is given by a patent to its proprietor.

Infringement is not determined by comparing the defendant’s allegedly infringing product to the Plaintiff’s product which is representative of the patent. Rather, the appropriate means of determining whether there is infringement is to compare the allegedly infringing product to the patent and, in particular, to the patent claims and

this rotates on construction of the claims.

Construction of claim and specification

As the specification is addressed to the person skilled in the art, the court must be instructed as to the meaning of technical terms and the background of the art, so as to read the specification with



the understanding of the person skilled in the art. The question of construction is necessarily antecedent to the determination of the issue of infringement and of all issues of validity depending upon the scope of the claims.

The onus of proving infringement lies on the patentee/plaintiff

Infringement is generally a question of fact (as *per Whirlpool Corp. v. Camco Inc., [2000] 2 S.C.R. 1067*) and a matter of construction of the claims. In establishing whether there is infringement or not two approaches are employed

The literal approach

In the older cases, if the words of the claim, upon a plain reading of them, aptly described the defendant's device or process, then there was said to be "literal" or "textual" infringement. Under literal infringement, in effect, all claim elements were considered to be essential to the invention and thus, the absence of any claimed element resulted in a finding of no literal infringement. It was therefore an "all elements" test. All the claim elements were considered important because it was the claim that determined the scope of the monopoly:

"In our view, the rule is that the claims must be regarded as definitely determining the scope of the monopoly, having regard to the due and proper construction of the expressions they contain." One of the earliest and most often cited case on claim construction relating to literal infringement is *Electric and Musical Industries, Ltd. et al v. Lissen, Ltd*, where Lord Russell of Killowen held at p. 39:

"The function of the claims is to define clearly and with precision the monopoly claimed, so that others may know the exact boundaries of the area within which they will be trespassers. Their primary object is to limit and not to extend the monopoly. What is not claimed is disclaimed. The claims must undoubtedly be read as part of the entire document, and not as a separate document; but the forbidden field must be found in the language of the claims and not elsewhere. It is not permissible, in my opinion, by reference to some language used in the earlier part of the specification to change a claim which by its own language is a claim for one subject-matter into a

claim for another and a different subject-matter, which is what you do when you alter the boundaries of the forbidden territory. A patentee who describes an invention in the body of a specification obtains no monopoly unless it is claimed in the claims. As Lord Cairns said, there is



no such thing as infringement of the equity of a patent. Also see (*Dudgeon v. Thomson, L.R. 3 App. Cas. 34*)."

This approach however, seems to have been so restricted and subsequently new methods were employed.

The purposive approach of construction

Purposive construction does not ask "What did the patentee say?" which might be a literal interpretation, but rather, "What was the inventor trying to say?" or, better yet, "What would a skilled reader have understood the patent to mean?" Section 42 of the Industrial Property Act states that the scope of protection available under the patent is to be determined by the terms of the claims, but where these are inadequate, the drawings and the description included in the patent may be used to interpret the claims.

The primacy of the language of the claims was emphatically affirmed in the celebrated case of *Catnic Components Ltd. v. Hill & Smith Ltd., [1982] R.P.C. 183 (H.L.)*. Lord Diplock summarized the test as follows:

"A patent specification should be given a purposive construction rather than a purely literal one", where "purposive construction" means understanding language in accordance with what a reasonable person would understand the author to be using the words to mean.

The tests to be used were aptly described in *Catnic Components Ltd. v. Hill & Smith Ltd. (1982) R.P.C. 183 (H.L. per Lord Diplock) at p. 242-243*:

"My Lords, a patent specification is a unilateral statement by the patentee, in words of his own choosing, addressed to those likely to have a practical interest in the subject matter of his invention (i.e. "skilled in the art"), by which he informs them what he claims to be the essential features of the new product or process for which the letters patent grants him a monopoly. It is called "pith and marrow" of the claim. A patent specification should be given a purposive construction rather than a purely literal one

derived from applying to it the kind of meticulous verbal analysis in which lawyers are too often tempted by their training to indulge...”

His Lordship stated the test as follows:



“...whether persons with practical knowledge and experience of the kind of work in which the invention was intended to be used, would understand that strict compliance with a particular descriptive word or phrase appearing in a claim was intended by the patentee to be an essential requirement of the invention so that any variant would fall outside the monopoly claimed, even though it could have no material effect upon the way the invention worked. The question, of course, does not arise where the variant would in fact have a material effect upon the way the invention worked. Nor does it arise unless at the date of publication of the specification it would be obvious to the informed reader that this was so. Where it is not obvious, in the light of then- existing knowledge, the reader is entitled to assume that the patentee thought at the time of the specification that he had good reason for limiting his monopoly so strictly and had intended to do so, even though subsequent work by him or others in the field of the invention might show the limitation to have been unnecessary. It is to be answered in the negative only when it would be apparent to any reader skilled in the art that a particular descriptive word or phrase used in a claim cannot have been intended by the patentee, who was also skilled in the art, to exclude minor variants which, to the knowledge of both him and the readers to whom the patent was addressed, could have no material effect upon the way in which the invention worked.”

The patent should be read as a whole

Mr. Justice Binnie endorsed a “purposive construction” to give the terms in the claim the meaning intended by the patentee based on a reading of the claims in the context of the patent as a whole. Regard may be had to the whole of the patent specification, including the drawings and the disclosure. This is consistent with section 42 of the *Industrial Property Act*.

Similarly, Taschereau J. stated in *Metallflex Ltd. v. Rodi & Wienenberger Aktiengesellschaft*:

"The claims, of course, must be construed with reference to the entire specifications, and the latter may therefore be considered in order to assist in apprehending and construing a claim, but the patentee may not be allowed to expand his monopoly

specifically expressed in the claims "by borrowing this or that gloss from other parts of the specifications".”

The principles that Lord Diplock offered in the Catnic case were summarized by Lord Hoffmann in *Improver Corporation v Remington Consumer Products Ltd* [1990] **FSR 181, 189** in terms of the three Improver principles or test procedures. Lord Hoffmann in that same decision



observed that a patentee may have intended a word or phrase to have not a literal but rather a figurative meaning, the figure being a form of synecdoche - (a form of the metaphor in which the part mentioned signifies the whole); or metonymy (a form of metaphor denoting the relation between two objects. Metonymy is to synecdoche what a metaphor is to a simile).

In *Kirin-Amgen Inc v Hoechst Marion Roussel Ltd* 8[2004] UKHL 46, [2005] 1 All ER 667, [2005] RPC 169 at 185, *Kirin-Amgen, Inc. ("Amgen")*, a Californian pharmaceutical company, was the proprietor of a patent relating to the production of the glycoprotein hormone erythropoietin (EPO) by recombinant DNA technology. Amgen sued Transkaryotic Therapies, Inc. ("TKT") for patent infringement. TKT, a Massachusetts corporation, had also developed a method of making EPO called GA-EPO using a process of gene activation. Hoechst Marion Roussel Ltd ("Hoechst") was sued for proposing to import TKT's GA-EPO into the United Kingdom. EPO, a previously known glycoprotein hormone, is made in the kidney and stimulates the production of red blood cells by the bone marrow. The essential difference between EPO and GA-EPO (for TKT) is that the former is made by an exogenous DNA sequence coding for EPO which has been introduced into a host cell and the latter is made by an endogenous DNA sequence coding for EPO in a human cell into which an exogenous upstream control sequence has been inserted.

The principal issue in the case was whether TKT's EPO fell outside the claims of Amgen's patent suit because of the difference in the way it was made. Lord Hoffman observed: Interpreting the claims of the Amgen patent in the context of its specification, the claims are concerned with the expression of EPO by a gene which is exogenous to the cell. But the genes which express EPO in cells by the TKT process are not exogenous. They come into existence when the cell is formed by division and simply replicate the pre-existing genes already present in the TKT cells. The TKT process works by a technique not generally known at the time when the Amgen patent application was published.

Lord Hoffman stated that: A claim may, upon its proper construction, cover products

or processes which involve the use of technology unknown at the time the claim was drafted. The question is whether the person skilled in the art would understand the description in a way which was sufficiently general to include the new technology. Lord Hoffmann concluded that that TKT did not infringe any of the claims and dismissed Amgen's appeal. The other lords all agreed.



Consequences of the decision: This case affirmed that: "Construction is objective in the sense that it is concerned with what a reasonable person to whom the utterance was addressed would have understood the author to be using the words to mean. Lord Hoffman cautioned: "Note, however, that it is not, as is sometimes said, "the meaning of the words the author used", but rather what the notional addressee would have understood the author to mean by using those words." (Paragraph 32 of the judgment)

NOTE:

The Catnic decision established the "Catnic principle":

- (a) The principle of purposive construction,
- (b) The guidelines for applying that principle to equivalents.

Forms of infringing activity

Case law illustrating infringement as envisioned by section 38 of the Industrial Property Act deals separately with the issues of

- a. Making and constructing;
- b. Using;
- c. Selling; and
- d. Importing.

Making and constructing

The manufacture of a patented product during the term of a patent is an infringement of the patent. It is not an infringement to sell non-patented components which later find themselves assembled into an infringing combination unless the vendor induced or procured the assembly of the components. However, the manufacture of all of the components that are later sufficiently assembled to test the fitting of the parts constitutes "making" a patented invention. This is the case even if the components are then disassembled and exported for assembly and use abroad.

A patent licensee may infringe by engaging in activities that are outside the scope of the licence. Where a patent claims a product for a particular use, it does not matter where that product is to be used, or even whether it is ever used. For instance, once the product is manufactured in



Canada for that use, there is infringement. Moreover, where a patent relates to a process for the manufacture of a product showing novel features, the product is presumed to have been manufactured by that process in the absence of proof to the contrary.

Using

Section 38 provides the exclusive right, among other things, to use the invention. Determining the meaning of "use" under s. 38 is essentially a matter of statutory construction. The Concise Oxford Dictionary defines "use" as "cause to act or serve for a purpose; bring into service; avail oneself of". First, the inquiry into the meaning of "use" under the Patent Act must be purposive, grounded in an understanding of the reasons for which patent protection is accorded. Second, the inquiry must be contextual, giving consideration to the other words of the provision. Finally, the inquiry must be attentive to the wisdom of the case law.

Identifying whether there has been infringement by use, like construing the claim, must be approached by the route of purposive construction: *Free World Trust v. Électro Santé Inc.*, 4 [2000] 2 S.C.R. 1024, 2000 SCC 66. "Purposive construction is capable of expanding or limiting a literal [textual claim]". The purpose of section. 38 IPA is to define the exclusive rights granted to the patent holder. These rights are the rights to full enjoyment of the monopoly granted by the patent. Therefore, what is prohibited is "any act that interferes with the full enjoyment of the monopoly granted to the patentee": Applied to "use", the question becomes: did the defendant's activity deprive the inventor in whole or in part, directly or indirectly, of full enjoyment of the monopoly conferred by law?

A purposive approach is complemented by a contextual examination of section. 38 Industrial Property Act, which shows that the patentee's monopoly generally protects its business interests. In determining whether the defendant "used" the patented invention, one compares the object of the patent with what the defendant did and asks whether the defendant's actions involved that object. In *Betts v. Neilson*, (1868),

L.R. 3 Ch. App. 429 (aff'd (1871), L.R. 5 H.L. 1 the object of the patent was to preserve the contents of bottles in transit. Though the bottles were merely shipped unopened through England, the defendant was held to have used the invention in England because, during its passage through that country, the beer was protected by the invention. Lord Chelmsford said, at p. 439:



“It is the employment of the machine or the article for the purpose for which it was designed which constitutes its active use; and whether the capsules were intended for ornament, or for protection of the contents of the bottles upon which they were placed, the whole time they were in England they may be correctly said to be in active use for the very objects for which they were placed upon the bottles by the vendors.”

The patented invention need not be deployed precisely for its intended purpose in order for its object to be involved in the defendant's activity. It was not relevant in Neilson whether the invention had actually caused bottles to be preserved during shipping, in a situation in which they would otherwise have broken. As a further example, in *Dunlop Pneumatic Tyre Co. v. British and Colonial Motor Car Co.* 9 (1901), 18 R.P.C. 313 (H.C.J.), the defendants placed on display at a car show a car with patented tires which they had intended to remove prior to sale, substituting other tires. The exhibition of the car with the patented tires was nonetheless held to be an infringing use. The common thread is that the defendants employed the invention to their advantage, depriving the inventor of the full enjoyment of the monopoly.

Monsanto Decision

(Monsanto Canada Inc. v. Schmeiser, [2004] 1 S.C.R. 902).

The Supreme Court of Canada discussed the law as it pertains to “use” in the Monsanto decision. In that case, Monsanto sued Schmeiser, a farmer, for infringement of a patent concerning herbicide-resistant canola. The claims in the patent related to a gene to make the canola herbicide-resistant, as well as a cell containing the gene. By saving seed from canola plants that had survived spraying with herbicide, Schmeiser produced a crop of over 1000 acres comprising over 95% patented canola. Based on these facts, the Court ruled that Schmeiser knew or should have known that the crop contained the patented cell or gene. Schmeiser defended that he had not infringed the patent because he had not made, constructed or used the patented cell or gene, or sold it to others to be used. He had simply gathered seeds from successful plants for planting the following year, a practice he had followed for

years.

The 5-4 majority of the Court was inclined to agree that Schmeiser did not make or construct the patented cell or gene. The main area of debate was whether Schmeiser had used the patented cell or gene. Schmeiser argued that he had not infringed that patent because the claims concerned a



cell or gene, whereas he had cultivated the whole plant. The plant itself could not be, and was not, patented. Schmeiser also argued that, because he had not sprayed herbicide on the crop, he had not exploited the advantage of the invention and thus had not used it. The Court offered the following list of propositions concerning the meaning of “use” (or its French translation “exploiter”);

1. “Use” or “exploiter”, in their ordinary dictionary meaning, denote utilization with a view to production or advantage.
2. The basic principle in determining whether the defendant has “used” a patented invention is whether the inventor has been deprived, in whole or in part, directly or indirectly, of the full enjoyment of the monopoly conferred by the patent.
3. If there is a commercial benefit to be derived from the invention, it belongs to the patent holder.
4. It is no bar to a finding of infringement that the patented object or process is a part of or composes a broader unpatented structure or process, provided the patented invention is significant or important to the defendant’s activities that involve the unpatented structure.
5. Possession of a patented object or an object incorporating a patented feature may constitute “use” of the object’s stand-by or insurance utility and thus constitute infringement.
6. Possession, at least in commercial circumstances, raises a rebuttable presumption of “use”.
7. While intention is generally irrelevant to determining whether there has been “use” and hence infringement, the absence of intention to employ or gain any advantage from the invention may be relevant to rebutting the presumption of use raised by possession.

The majority of the Court dismissed the first non-infringement defense (that Schmeiser cultivated the plant, not the patented cell or gene contained therein), stating as follows: “Provided the patented invention is a significant aspect of the

defendant's activity, the defendant will be held to have "used" the invention and violated the patent... Infringement does not require use of the gene or cell in isolation. "With regard to the second non-infringement argument (that Schmeiser had not sprayed herbicide on his crop and thus had not exploited the invention), the majority of the Court ruled that the argument failed to take into account the stand-by or insurance



utility of the properties of the patented genes or cells. Schmeiser benefited from the invention by virtue of the fact that he could have sprayed herbicide if the need had arisen. He did not provide sufficient evidence to rebut the presumption of use that arose from his possession of the crop containing the patented genes and cells.

Where a patent claims a product for a particular use, it does not matter where that product is to be used, or even whether it is ever used. Once the product is manufactured for instance in Canada for that use, there is infringement.

In discussing the presumption that possession constitutes use, the majority noted the general rule that the defendant's intention is irrelevant to a finding of infringement. The majority also discussed examples in which the "use" of an invention would constitute exploitation of the invention's stand-by utility. A fire extinguisher is "used" to provide the means for extinguishment should the need arise. This is so even if there is never a fire to be extinguished. Patented pumps on a ship were "used" even though they were never used to pump water in British waters, because there was an intention to use them if the need arose.

Other examples include:

Manufacture

The manufacture of all of the components that are later sufficiently assembled to test the fitting of the parts constitutes "making" a patented invention for the purposes of section 38 IPA. This is the case even if the components are then disassembled and exported for assembly and use abroad.

Where infringing goods pass down a chain of distributors, each person becomes liable as keeper, disposer or distributor. See *British Shoe Manufacturers V Collier (1910)* 27 *RPC* 567

A person who acquires or imports goods with a view of exporting them/sell abroad will be deemed to infringe. See S. 38(1)(b) of the **Industrial Property Act 2014**.

How about sale of independent elements of an invention and not the whole of it,

does it amount to infringement? In *Beloit Canada Ltd. v. Valmet-Dominion Inc. (1997)*, 73 C.P.R. (3d) 321 (F.C.A.), the Federal Court of Appeal confirmed at p. 337 that “where the elements of an invention are sold in a substantially unified and combined form for the purpose of later assembly,



infringement may not be avoided by a separation or division of parts which leaves to the purchaser a simple task of integration and assembly”.

The exclusive rights of a patent last only during the term of the patent. For example, the manufacture of an article during the term of a patent is an infringement, but the use or sale of that article after the monopoly comes to an end is not.

Stocking the product for the purposes of offering for sale.

Note however, that a mere carrier is not a keeper for purposes of the Act. In ***Smith Kline & French V Harbottle***[1980] **RPC 363** the 1st defendant, Harbottle ordered a drug patented by the plaintiff in U.K, from Italy. It arranged for its importation into England with the intention of re- exportation. It was carried by British Airways and stored there in their bonded warehouse to the order of Harbottle. The plaintiff alleged that, British Airways thereby infringed the patent as a person who keeps the product whether for disposal or otherwise. *Olive J* while holding that British Airways did not infringe the patent by carrying the goods or warehousing them, interpreted keeping as: “...What the draftsman had in his mind was keeping in the sense of keeping in stock so as to give effect to the words of the Convention “stocking the product for these purposes” and while referring to the Shorter Oxford English Dictionary as to the means of the verb’ to stock...as meaning to keep goods in stock for sale”

Repairs

The owner of a patented article is entitled to make genuine repairs (These repairs must not, however, amount to making a substantially new article: see ***Dunlop Pneumatic Tyre Co Ltd v Neal***), or to replace parts of a combination which normally wear out more quickly than other parts, even if, in so doing, he would otherwise infringe any copyright belonging to the proprietor of the patent. The implied licence to repair extends to the purchaser's agents and contractors. See ***Solar Thomson Engineering Co Ltd v Barton*** [1977] **RPC 537, CA**.

However, this implied license does not apply where the work goes beyond repair and

amounts to remanufacture of the patented article resulting in a new merchantable article. Such remanufacture constitutes infringement of the patentee's exclusive right to 'make' or 'construct'



the invention. Whether work performed on a patented article constitutes repair or amounts to remanufacture is a question of fact that will depend on the nature of the patented article and the character of the work carried out upon it. For example, in the case of *Rucker Co. v. Gavel's Vulcanizing Ltd. (1985)*, 7 C.P.R. (3d) 294, [1985] F.C.J. No. 1031 (F.C.T.D.) at p. 325, it has been held that, when a manufacturer takes the whole thing and sells what is a new tire with merely the old wires in it, there has been no license to use those old wires for the purpose of putting them into and making up precisely the same combination which is the subject of the patent.

The doctrine of variants

This means that the court decides whether the defendant was employing the patentee's inventive concept through some measure of equivalence to what had actually been claimed. It has long been established that if a person takes the substance of an invention he is guilty of infringement even if his act does not in every respect fall within the express terms of the claim defining it. This basic principle was stated as early as 1875 by James L.J. in *Clark v. Adie (1875)*, L.R. 10 Ch. 667 at p. 675 in the following terms:

“A patent for a new combination or arrangement is to be entitled to the same protection, and on the same principles, as every other patent. In fact, every, or almost every, patent is a patent for a new combination. The patent is for the entire combination, but there is, or may be, an essence or substance of the invention underlying the mere accident of form; and that invention, like every other invention, may be pirated by a theft in a disguised or mutilated form, and it will be in every case a question of fact Whether the alleged piracy is the same in substance and effect, or is a substantially new or different combination.”

When *Clark v. Adie went to the House of Lords (1877)*, 2 App. Cas. 315, the Lord Chancellor (Lord Cairns) discussed the various ways in which a patent for an apparatus could be infringed. In the course of his discussion he said, at p. 320:

“The infringer might not take the whole of the instrument here described, but he might take a certain number of parts of the instrument described; he might make an instrument which in many respects would resemble the patent instrument, but would not resemble it in all its parts. And there the question would be, ..., whether that which was done by the alleged infringer amounted



to a colourable departure from the instrument patented, and whether in what he had done he had not really taken and adopted the substance of the instrument patented. And it might well be, that if the instrument patented consisted of twelve different steps, ..., an infringer who took eight or nine or ten of those steps might be held by the tribunal judging of the patent to have taken in substance the pith and marrow of the invention, although there were one, two, three, four or five steps which he might not actually have taken and represented upon his machine.”

Lord Cairns appears to have been the originator of the mixed metaphor “the pith and marrow of the invention”. While the metaphor has been criticized the principle enunciated in *Clark v. Adie*, *supra*, has been followed and applied in many cases, both in Great Britain and in Canada, and has never been repudiated: vide, for example

1. *Proctor v. Bennis* (1887), 4 R.P.C. 333 at pp. 345, 352, 362;
2. *Benno Jaffe und Darmstaedter Lanolin Fabrik v. John Richardson & Co. (Leicester) Ltd.* (1893), 11 R.P.C. 93 at pp. 112; 261;
3. *Incandescent Gas Light Co. V. De Mare Incandescent Gas Light System Ltd.* (1896), 13R.P.C. 301 at p. 331; 559 at pp. 571, 579;
4. *Marconi v. Br. Radio Telegraph & Telephone Co.* (1911), 28 R.P.C. 181 at p. 217;
5. *Br. ThomsonHouston Co. V. Metropolitan-Vickers Electrical Co.* (1928), 45 R.P.C. 1 at p. 25;
6. *Rheostatic Co. v. Robt. McLaren & Co.* (1935), 53 R.P.C. 109 at p. 118;
7. *Lightning Fastener Co. V. Colonial Co.* [1932] Ex. C.R. 89 at pp. 98, 100; [1934] 3 D.L.R. 737 at pp. 743-4, 51.

In *Clark v. Adie*, Lord Cairns did not specifically refer to the doctrine of mechanical equivalency but it is implied in his statement. Indeed, it is only a particular application of the general doctrine enunciated by him. That this is so was stated by Romer L.J. in *R.C.A. Photophone Ltd. v. Gaumont-Br. Picture Corp. & Br. Acoustic Films Ltd.* (1936), 53 R.P.C. 167 at p. 197; where he said of it:

“The principle is, indeed, no more than a particular application of the more general principle that a person who takes what in the familiar, though oddly mixed metaphor is called the pith and marrow of the invention is an infringer. If he takes the pith and marrow of the invention, he commits an infringement even though he omits an unessential part. So too, he commits an



infringement if, instead of omitting an unessential part, he substitutes for that part a mechanical equivalent.”

To determine whether a claim element is essential or not, ask whether

- It is obvious to the skilled reader at the time the patent was published that a variant of a particular element would not make a difference to the way in which the invention works? If modifying or omitting the element changes the way the invention works, and that was obvious at the relevant date, then the element is essential;
- According to the intent of the inventor, expressed or inferred from the claims, was a particular element essential irrespective of its practical effect? If the element appears to have been intended to be essential, then the element is essential.

If the claim element was not found to be essential under either Q1 or Q2, then the element is non-essential.

How to determine infringement by a variant?

This can be resolved by considering whether the differences between the variant and the patented invention are material, that is, do they differ in essential or inessential respects. The invention claimed can be considered as comprising essential and nonessential integers (components), those that are fundamental to the invention and those that are not. If the alleged infringer has taken all of the essential integers, then there is infringement even if there are substantial differences in respect of the nonessential integers. 65 This has been described by various judges as the “pith and marrow” of the invention. In applying this principle, the scope of the patent claims is vital. *Viscount Radcliff said in Van Der Lely NV V Bamford Ltd {1963} RPC 61 at 78:* When.... the pith and marrow principle is invoked to support the accusation; one must be very careful to see that the inventor has not by the actual form of his claim left open to the world the appropriation of just that property that he says has been filched from him...”

Thus, an inventor must be very careful when drafting his claims to make sure that

they are not framed too narrowly that some slight and insignificant modification can be affected without infringing the patent.



The question of construction of patent claims in the context of variants was considered *in Catnic Components Ltd V Hill & Smith Ltd {1982} RPC 183*, which involved several variants of steel lintels. The claimant was a proprietor of a patent for steel lintels that had a rear support member which was vertical and so described in the claim of the specification by the phrase ‘second rigid support member extending vertically from or from near the rear edge of the first horizontal plate or part adjacent its rear edge’. The defendant made a similar lintel, but with the rear support member inclined between six and eight degrees (depending on the particular mode of lintel) from the vertical. The defendant’s lintel had a reduced load-bearing capacity compared to the claimant’s lintel, but because of the small inclination from the vertical this reduction was small. The HOL found the claimant’s patent had been infringed and the defendant’s argument that the verticality of the claimant’s lintel was essential to its function and that, therefore, there was no pith and marrow infringement was rejected. It was confirmed that a purposive approach should be adopted in the construction of the patent specification. Lord **Diplock** *in Catnic Components Ltd V Hill & Smith Ltd (at 243)* stated the test as being whether practical persons skilled in the art, would understand that strict compliance with a particular word or phrase was intended by the patentee to be an essential requirement of the invention. If so, any variant that did not comply would fall outside the claim regardless of whether it had any effect, if the variant did have a material effect, there would be no infringement. Lord Diplock suggested the test for deciding whether variants infringe later adopted by Hoffman J *in Improver Corp V Remington Consumer Products Ltd {1990} FSR 181 at 189*. The three-part test is as follows;

- i. Does the variant have a material effect on the way the invention works? If yes, the variant is outside the claim (and does not infringe). If no
- ii. Would this (i.e that the variant had no material effect) have been obvious at the date of plaction of the patent to a reader skilled in the art? If no, the variant is outside the claim. If yes-
- iii. Would the reader skilled in the art nevertheless have understood the language of

the claim that the patentee intended strict compliance with the primary meaning was an essential requirement of the invention? If yes, the variant is outside the claim. (Can be expressed as: whether a skilled reader would understand from the language of the claim that strict compliance with the primary meaning of the claim was intended).



On the other hand, a negative answer to the last question would lead to the conclusion that the patentee was intending the word or phrase to have not a literal but a figurative meaning denoting a class of things which included the variant and the literal meaning, the latter being perhaps the most perfect, best –known or striking example of the class.

NB. The first 2 questions in **Improver case** are qns of fact and the 3rd is based on interpretation.

The purposive approach and the Catnic case have become the basic methods of determining whether variants infringe.

DOCTRINE OF EQUIVALENTS

Sometimes the alleged infringement may contain mechanical equivalents, an alternative that works equivalently. A mechanical equivalent may be more than a mere variant as in the Catnic case. It is then a question of deciding which are the essential integers of the claim in the pith and marrow approach to interpretation discussed above. E.g. in *Rodi & Wienberger AG V Henry Showell Ltd [1969] RPC 367* it was held that replacing two U-shaped bows in a flexible watch strap with a single large C-shaped bow was not an infringement because the U-shaped bow was an essential integer. Alternatively, in *Marconi V British Radio Telegraph & Telephone [1991] FSR 223* the replacement of an auto-transformer with a two-coil transformer did not prevent a finding of infringement because the auto-transformer was not an essential integer.

DEFENSES TO INFRINGEMENT

1. Prior use and commencement of infringement

This is provided for under section 41 of the Industrial Property Act of 2014 where if a person uses an invention in a way that makes it public before the priority date of a patent for it, his anticipation will render the patent invalid. The infringer has a defense

against infringement should he continue to use the invention after the patent's right takes effect. The defense is also open to people who have not provided material for attacking the patent's validity, prior users whose activities have not made the invention available to the public.

2. Delay and acquiescence



Where the patentee delays to bring a cause of action in patent infringement, it will be deemed that he/she has consented and will be estopped from claiming otherwise.

3. License

Section 50 of the Industrial Property Act provides for the rights of a licensee and section 65 provides for license as a right. Where one has been given a license to use a patent, he or she is protected from infringement.

4. Invalidity of a patent

Where a patent doesn't qualify with *sections 8,9,10,11,12* of the Industrial Property Act which provide for novelty, inventive step and industrial application, such a patent is invalid and in infringement proceedings it is proper for the alleged infringer to contend that the patent is invalid for want of satisfaction of any of the essential prerequisites of a patent.

5. Term of a patent

Section 46 of the Industrial Property Act provides that a patent shall expire at the end of twenty years from the filing date of the application therefor where a person uses such a patent after the expiration of twenty years, he or she has a defense against infringement.

REMEDIES UNDER INFRINGEMENT

The remedies for infringement are provided for under section 40 of the Industrial Property Act and they include the following;

Injunction.

Section 40(a) of Industrial Property Act provides that the owner of the patent has a right to obtain an injunction to restrain the performance or the likely performance by any person without his or her authorization of any act referred to in section 38 of the Industrial Property Act. In the case of

Victoria Park Racing V Taylor [1937] HCA 45, it was held that injunctions are

granted only on intangible elements value and that they are to protect to the exclusive right of the invention and other intellectual property rights.

Damages.



Section 40(b) Industrial Property Act provides that the owner of the patent with a right to claim damages from any person who having knowledge of the patent performs any act in section 38 of the IPA without the owner's authorization. In the case of *Watson Laidlaw V Potts Cassels [1914] UKHL 238 (06 February 1914)* it was held that it is the restoration principle that is used as an idea to restore the person who has sustained injury or loss to the condition in which he would have been had he not sustained it.

Compensation.

Section 40(c) of Industrial Property Act provides that the owner of the patent has the right to claim compensation from any person who without his or her authorization performs any of the invention claimed in the published application as if a patent had been granted for that invention.

Equitable Remedies

Account for profits

The infringer can be ordered to account for the profits gained as a result of infringement and the principles for accounting for the profits of infringement were laid down in the case of Celanese *International Co-Operation V BP Chemicals Ltd [1999] RPC 203*, where it was held that it depends on the legal causation, whether the defendant carries out several activities or sells different products.

Anton pillar order.

This was laid down in the case of *Anton Piller K.G V Manufacturing Processes Ltd CA 8 Dec 1975*. And it gives the right to the owner of the patent to search the premises and seizure all evidence without prior warning of the right.

THE PROCEDURE FOR OBTAINING A PATENT (SECTION 21 IPA & REGULATION 6)

Step 1.

Filing of patent application with the Registrar

Application fee (Shs. 180.000/-) (Section 21 (1)

IPA 2014

Step 2.



Formal and substantive examination of the application is carried out by the patent office upon payment of prescribed fees. A formal examination is done by the Registrar of patents and then forwarded to the African Regional Industrial Property Organization (ARIPO) for substantive examination. The application is published in the Gazette.

(Section 28, 29, 30 & 31 IPA 2014 and Regulations 27 & 28 of the Patent Regulations SI 216-1) Refer to the Industrial Property Regulation 2017

Step 3.

A certificate of grant and a copy of the granted patent are issued by the intellectual property office upon a finding that the application has patentable subject matter, payment of the prescribed fees has been made, and that there is no opposition application or such opposition application has been unsuccessful.

Grant fee (Shs. 300.000/-)

Step 4.

Publication of patent in the Uganda Gazette

(Section 32 IPA)

REGIONAL PROTECTION

This entails patent protection across countries or states that are parties to the relevant various patent treaties.

The protection can be acquired by obtaining a patent under which is authorized by the Harare protocol of 1982 which empowered the ARIPO secretarial to register and administer patents on behalf of member states.

ARIPO is mandated to grant patents on behalf of the Harare Protocol Contracting States in accordance with the provisions of the Harare Protocol on Patents, Designs

and Utility Models. The Harare Protocol was adopted on December 10, 1982 at Harare, Zimbabwe. The protocol entered into force in 1984.



Under the protocol, an applicant for the grant of a patent for an invention can, by filing only one application, designate any one of the Harare Protocol Contracting States in which that applicant wishes the invention to be accorded protection. The protocol requires the filing of the application to be made with any one of the Contracting States or directly with the ARIPO Office.

There are 18 Contracting States, namely Botswana, Kingdom of Eswatini, The Gambia, Ghana, Kenya, Kingdom of Lesotho, Liberia, Malawi, Mozambique, Namibia, Rwanda, Sao Tome and Principe, Sierra Leone, Sudan, Tanzania, Uganda, Zambia and Zimbabwe.

Section 45 Industrial Property Act, 2014 provides for the Harare protocol on patent, that a patent in respect of which Uganda is a designated state, granted by ARIPO by virtue of the Harare protocol has the same effect in Uganda as a patent granted under this Act except where the registrar communicates to ARIPO in respect of the application of the patent, a decision in accordance with the provisions of protocol that if a patent is granted by ARIPO, that patent shall have no effect in Uganda.

Procedures for Filing of a Patent

Filing an Application

Applications may be filed by the applicant (who can be an inventor or his/her assignee) or by the authorized representative of the applicant (attorney, agent, or legal practitioner) who has the right to represent the applicant before an industrial property office of any of the Harare Protocol contracting states. Representation is mandatory for applicants who are not resident or whose principal place of business is not situated in a contracting state. To File the Applicant has to submit the following:

- A completed ARIPO Form 3 (Request form)
- A description of the invention
- One or more claims
- One or more drawings (if any)
- An abstract

- Prescribed application fees or a written undertaking to lodge the fees
- Designation of at least one state



- The application is filled with the industrial property office in Uganda which transmits it to ARIPO upon payment to the requisite fee.
- Alternatively, one may apply straight to the ARIPO secretariat in Harare on payment of the requisite fee.
- Under the protocol an applicant for the grant of a patent or the registration of an industrial design can by filling only one application, designate any of the contracting states in which he wishes his invention or industrial design to be accorded protection.
- The protocol requires the filling of the application to be made with either one of the contracting states or directly with the ARIPO office.
- The ARIPO office, on the receipt of the patent application, carries out the substantive examination to ensure that the invention which is the subject of the application, is patentable, in other words that is new, involves an inventive step and is capable of being applied in the industry.
- When the application complies with the substantive requirements, the application is advertised in the ARIPO member states and copies thereof are sent to each designated contracting states.
- The said countries must be notified to either accept or reject the patent.
- In case of any objection, they must be communicated to ARIPO within 6 months indicating the grounds as to why it is not patentable under its domestic law to the ARIPO office that according to grounds specified in the protocol should ARIPO grant the patent, that grant will not have effect in its territory.
- The substantive examination of ARIPO patents particularly a story one and the examination substantially raises its presumption of validity.
- After granting the patent, fees are paid to ARIPO office which in turn sends the same or may require the applicant prosecute the application in designated companies.
- The patent once granted is as good as a patent by a member state.

PROCEDURE FOR PATENT APPLICATION UNDER
INTERNATIONAL PROTECTION



Patent application is a request pending at a patent office for grant of a patent for an invention described in the patent specification and a set of one or more claims stated in a formal document including necessary official form related correspondences.

To obtain a grant of patent, a person either legal or national, must observe the following;

1. File a PCT application

This is to the effect that one must file an application at the patent office with jurisdiction to a grant a patent in a geographical area over which coverage is required. This is usually at the receiving office defined under Article 2 PCT treaty to mean the national office of intergovernmental organisation with which the international application has been filed or may also be a regional body like European patent office.

In regards to the above, the Application may be filed with ARIPO or international bureau of the WIPO.

2. Conduct of search

This is provided for under Art: 15 of the PCT treaty which provides for an international search by the International Search Authority under Art: 16 (ISA), accompanied by a written opinion regarding patentability of the invention which is the subject matter of the application.

3. Carrying out international preliminary examination

Here a preliminary examination must be carried out by an International Preliminary Examination Authority (IPEA). This is provided for under Art(s): (31, 32, 33, 34, 35 and 36) of the PCT treaty. In regards to the above, ARIPO carries out the formal examination and substantive examination of the invention and forwards it together with a accompanying document to WIPO.

4. Acquisition of a local attorney

During the nationalization stage, one must file proper work in the state/country of

their choice and in most cases one will need a local attorney to handle proper work or prosecute his patent.

5. Payment of fees

In most patent laws, renewal annuities or maintenance fees have to be regularly paid in order to keep the patent in force.



Otherwise patent lapses by judgement of a court, as where a claim or patent is held invalid under the relevant law thus no longer enforceable. However, it should be noted that patent fees for each state or country refers depending on the type of invention.

6. Grant of patent

Once the patent specification complies with the laws of the office concerned, a patent may be granted for the invention described and claimed by the specification. This exercise is usually done by the national or regional authority which administers matters related to examination of application, thus the patent can be granted to the member states.

RELATIONSHIP BETWEEN WIPO AND ARIPO

1) The signing of the memorandum of understanding between ARIPO and WIPO.

The director general of the WIPO, Mr. Francis Gury and his counterpart from ARIPO, Mr. Gift H Sibanda signed memorandum of understanding on Oct 2nd, 2008 in Geneva to, strengthen and expand cooperation between the organizations.

The M.O.U cover traditional areas of cooperation relating to capacity building as well as special projects to boost ARIPO's patent information capacity.

2) Collaboration between WIPO and ARIPO

In addition to re-enforcement, WIPO and ARIPO's 30-year-old collaboration, Mr. Gury said today's agreement will boost the capacity of ARIPO member states to provide the means for research institutes, universities and private sector to capitalize on the wealth of technical information available in patent documents for example, university Dar-el-salaam now offers a bachelor's degree in patent which is an achievement as a result of WIPO and ARIPO's collaboration.

3) Improvement on access to efficient use of technology Gury noted that improving access to efficient use of technological information contained in patent documents is

crucial and WIPO will provide support to ARIPO and its member states to broaden their ability to participate more fully in the knowledge of economy.



4) Assistance in undertaking studies on the construction of copy right based industries. WIPO will also assist, where appropriate in undertaking studies on the contribution of copy right based industries to national economies and the role of distinctive signs, particularly geographical indications in the promotion of domestic and export trade. This will further extend to building of capacity of copy right collective management organization and enhancing rights management at national or regional level.

Provision of technical and legislative assistance.

This is to the effect that, cooperation will also be strengthened in establishing a legal frame work for the protection of traditional knowledge and expression of folklore. This is stipulated in the Swakopmund protocol signed in 2010 and came into force on 11th May, 2015.

5) Provision of technical assistance to digitize and format patent data.

In a separate agreement relating to patent documents, WIPO will provide ARIPO with necessary technical assistance to digitalize and format patent data. This will enable ARIPO to provide value added patent information services to its member states.

6) Development of patent information tools

WIPO and ARIPO will cooperate in the development of patent information tools to demonstrate the value of patent information and to increase awareness about the utilization of patent information.

7) Integration of patent information

The two organs will jointly organize symposia, Seminars, workshops and other training programs in the area of patent information to address a particular needs and requirement of users from ARIPO member countries.

Further, WIPO will integrate the patent information provided by ARIPO and its

member states into WIPO'S search service to encourage the broader dissemination and use of data.

8) Harmonization of intellectual property procedures

ARIPO established in 1976, deals with the harmonization of the intellectual procedures, awareness building and administration of intellectual property rights among its 18 member states



such as Botswana, Kingdom of Eswatini, The Gambia, Ghana, Kenya, Kingdom of Lesotho, Liberia, Malawi, Mozambique, Namibia, Rwanda, Sao Tome and Principe, Sierra Leone, Sudan, Tanzania, Uganda, Zambia and Zimbabwe.

